MILITARY SPECIFICATION

CLAMPS, INSTRUMENT MOUNTING, AIRCRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers aircraft instrument mounting clamps.

* 1.2 Classification. The clamps shall be furnished in the following nominal sizes, as specified (see 6.2):

Clamp Size (Inches)	Clamp MS Part No.
1	MS28042-4A
1-1/2	MS28042-3A
2	MS28042-1A
3	MS28042-2A

2. APPLCIABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-200/3	Aluminum Alloy Bar, Rod, Shapes, Tube, And Wire, Extruded 2024
QQ-A-200/8	Aluminum Alloy Bar, Rod, Shapes, Tube, And Wire , Extruded And Structural Shapes, 6061
QQ-A-250/4	Aluminum Alloy 2024, Plate And Sheet
QQ-A-250/11	Aluminum Alloy 6061, Plate And Sheet
QQ-A-430	Aluminum Alloy Rod And Wire, For Rivets And Cold Heading
PPP-H-1581	Hardware (Fasteners And Related Items) Packaging And Packing For Shipment And Storage Of

Military

MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-C-6390	Clamps And Instruments, Aircraft, Installation Of

FSC 5340

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:30Z
Check the source to verify that this is the current version before use.

MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-L-46010	Lubricant Solid Film, Heat Cured Corrosion Inhibiting

STANDARDS

MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-810	Environmental Test Methods
MS24693	Screw Machine, Flat Countersunk Head, 100 Deg., Cross Recessed, UNC-2A And UNF-2A (IN/MM)
MS28042	Clamp, Mounting, Aircraft Instruments
MS33639	Cases, Instrument, Clamp Mounted, Aircraft
MS35214	Screw Machine Pan Head, Cross Recessed, Brass, Black Oxide, UNC-2A
MS35215	Screw Machine Pan Head, Cross Recessed, Brass, Black Oxide, UNF-2A

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Material. Clamps covered by this specification shall be manufactured per MS28042 from aluminum alloy 2024 plate and sheet specification QQ-A-250/4 or aluminum alloy 6061 plate and sheet specification QQ-A-250/11.

* 3.2 Design.

* 3.2.1 Function. Each clamp shall be designed to be used for mounting an instrument whose case conforms to MS33639 for its respective nominal size, when installed on an instrument panel of any thickness up to and including 1/4 inch in accordance with MIL-C-6390. Each clamp shall conform to MS28042. The clamp need not have the shape shown on MS28042, but must be contained within the outline shown on MS28042.

* 3.2.2 Roundness. The clamp shall be uniformly round when in the closed position. The inside surface of the clamp shall be clean, sound, smooth, and free from injurious imperfections.

* 3.2.3 Adjustability. The clamp shall be provided with a mechanism which will change the diameter of the clamp band within the limits specified on MS28042 when it is actuated by a screw or another suitable means which is operated from the front of the panel.

* 3.2.4 Limiting. The mechanism shall be provided with a limiting action which will be effective when the clamp is installed on a panel of any thickness up to and including 1/4 inch and is tightened to a point where an instrument case conforming to MS33639 is held securely in place.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:30Z
Check the source to verify that this is the current version before use.

* 3.2.5 Self-spreading. The clamp shall be self-spreading when the mechanism is loosened. The mechanism shall have no tendency to stick or bind when operated. Dry film lubrication conforming to MIL-L-46010 may be used to insure proper operation of the mechanism.

* 3.2.6 Disengagement. Each clamp shall be so designed that when the clamp is installed in the panel, no part of the clamp assembly, except the screw when one is used as an actuator, shall become disengaged from any other part by any operation of the actuator, whether or not an instrument is installed in the clamp.

* 3.2.7 Loosening. It shall be possible to remove an instrument whose case conforms to MS33639 from the front of the panel without any binding when the clamp is properly installed on the rear of the panel and the actuator is loosened or turned a maximum of four times.

* 3.2.8 Mounting lug. The mounting lug shall be provided with a self-locking feature.

* 3.2.8.1 Mounting screw. Mounting screws shall be selected from MS24693 and shall be brass, black oxide coated. Screws shall be of sufficient length to properly mount the clamp in panel 1/4 inch thick. Mounting screws shall be furnished as part of the clamp.

* 3.2.9 Actuator. The clamp shall be actuated by a screw conforming to 3.1.9.1. The actuator body shall be within the dimensional envelope specified on MS28042 and shall permit unrestricted tightening and loosening when clamp and instrument are mounted in panel 1/4 inch thick and of dimensions specified in MIL-C-6390.

* 3.2.9.1 Actuator screws. Actuator screws shall be selected from MS35214 and MS35215 except as noted. Screws shall be of sufficient length to permit complete loosening of the clamp when mounted in a 1/4 inch thick panel. Thread sizes shall be as follows:

Clamp Size (Inches)	Thread Size
1	6-32 UNC-2A
1-1/2	6-32 UNC-2A
2	10-32 UNC-2A
3	10-32 UNC-2A

The head diameter of screws for the 3 inch size clamp shall be .245 to .240 inches. Standard screws may be reworked to meet this requirement (and re-oxide coated) or new screws may be manufactured that meet all requirements of MS35214 but with the reduced head diameter.

* 3.3 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:30Z
Check the source to verify that this is the current version before use.

* 3.4 Weight. The weight of each clamp including its mounting and actuating screws shall not exceed the following value:

Clamp Size (Inches)	Maximum Weight (Ounces)
1	1.00
1-1/2	1.75
2	2.00
3	2.25

* 3.5 Finish. Aluminum-alloy parts shall be covered with an anodic film conforming to Type II of MIL-A-8625.

* 3.6 Performance. The clamp shall be capable of complying with the requirements of Section 4 when subjected to the applicable tests.

* 3.7 Identification of product. The clamp shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of clamps shall be classified as acceptance tests (4.3).

* 4.3 Acceptance tests. The acceptance tests shall consist of individual tests and sampling tests.

* 4.3.1 Individual tests. Each clamp shall be subjected to the following test:

Examination of product (4.5.1)

* 4.3.2 Sampling tests. The sampling tests shall consist of sampling plan A tests and sampling plan B tests.

* 4.3.2.1 Sampling plan A. Three percent of the clamps, but at least 12 clamps, shall be selected at random from each lot of clamps and shall be subjected to the following tests:

a. Operation (4.5.2).

b. Vibration (4.5.3).

* 4.3.2.2 Sampling plan B. When specified (see 6.2), 5 clamps selected at random from the first 12 items on the contract or order shall be subjected to the following test:

Humidity (4.5.4).

Source: https://assist.dla.mil -- Downloaded: 20260-17T07:30Z
Check the source to verify that this is the current version before use.

* 4.3.3 Lot. A lot shall consist of clamps of the same size, manufactured from like materials for shipment at one time.

* 4.3.4 Rejection and retest. The instructions on reworking or resubmitting rejected items shall be as specified in the general provisions of the contract.

* 4.3.5 Defects in items already accepted. If investigation of a test failure indicates that defects may exist in items already accepted, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

* 4.4 Test conditions.

* 4.4.1 Test instrument. Whenever a test instrument is mentioned, the case of the test instrument shall conform to MS33639. The test instrument case shall be made of metal. The wall thickness shall be at least 1/32 inch. The finish on the exterior of the test instrument case shall be durable dull black, and the exterior shall be uniformly smooth. The weight of the test instrument shall be the following value ±10 percent:

Clamp Size (Inches)	Test Instrument Weight (Pounds)
1	3/4
1-1/2	1
2	2
3	4

If it is necessary to add weights to the test instrument to obtain the desired weight, they shall be securely attached to the inside of the case. No tubing or wiring harness shall be connected to any test instrument.

* 4.4.2 Test panel. Whenever a test panel is specified, it shall be 1/8 inch thick fabricated from an aluminum alloy conforming to QQ-A-250/4, T3. The clamps shall be installed on the panel in accordance with MIL-C-6390. The inter-instrument distance shall not exceed by more than 1/32 inch the minimum value which MIL-C-6390 permits, and the clamps shall be installed in a square grid pattern as follows:

Clamp Size (Inches)	Test Panel Clamps Pattern
1	3 clamps high by 4 clamps wide.
1-1/2	3 clamps high by 4 clamps wide.
2	3 clamps high by 4 clamps wide.
3	2 clamps high by 3 clamps wide.

* 4.4.3 Vibration stand. Whenever a vibration stand is specified, it shall vibrate at a frequency of 300 to 3,000 cycles per minute, and shall subject the clamp to such vibration that a point on the clamp will describe, in a plane inclined 45 degrees from the horizontal, a circle of not less than 0.034 inch nor more than 0.036 inch in diameter.

* 4.5 Test methods.

* 4.5.1 Examination of product. Each clamp shall be examined to determine conformance to this specification as to workmanship, finish, dimensions, and materials. Except as specified herein, dimensional checks shall be made on

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:30Z
Check the source to verify that this is the current version before use.

unmounted clamps. The distance between the center of the clamp, shall be measured with the clamp firmly holding a case conforming to MS33639.

* 4.5.2 Operation. A clamp shall be installed in each cutout on the test panel. A test instrument shall be inserted in each clamp. The front surface of the test panel shall be checked for flatness on all four sides within 1/4 inch of the edge of the panel by use of a straightedge or other suitable means. The actuators of the clamps shall then be tightened until the test cases are held firmly in place. The test instrument cases shall have no tendency to cock. This shall be determined by examining the shoulder of the instrument to see that it is resting on the panel for its entire circumference. It shall not be possible to fully insert a feeler gage, 0.001 by 0.125 inch between the clamp and the instrument case for at least an integral 90 percent of the case periphery. The test panel shall again be checked for flatness as described above while the actuators are still tight. The panel shall not show deflection of more than 1/32 inch as a result of tightening all of the actuators. Each of the actuators shall then be loosened a maximum of four turns. The test instruments shall then be removed by gripping the shoulder at the front of the case and pulling the instrument forward. There shall be no evidence of binding between the clamps and the instruments during the removal process.

* 4.5.3 Vibration. A test panel complete with clamps and test instruments shall be installed on a vibration stand. The actuating mechanism and mounting screw of each clamp shall be indexed. The test panel shall then be subjected to the vibration test conditions specified in 4.4.3 for 6 hours. The frequency shall be varied exponentially such that an equal number of test cycles will accrue at each frequency. At the end of this test, each clamp shall be examined to determine that neither the actuator nor the mounting screw has turned from its index point and that the test instrument is still held firmly in place.

* 4.5.4 Humidity. The clamps shall be subjected to the humidity test in accordance with Procedure I of MIL-STD-810, but for only 10 cycles. At the conclusion of the test, the clamps shall be thoroughly examined. There shall be no evidence of corrosion.

* 4.6 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging, packing and marking. Clamps shall be packaged, packed and marked in accordance with PPP-H-1581. Levels of packaging shall be as specified by the procuring activity (see 6.2).

5.2 Marking of shipments. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:30Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The clamps covered by this specification are intended for use for mounting instruments whose cases conform to MS33639 in instrument panels in aircraft.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. MS part No. of the clamp desired (see 1.2).

c. Selection of applicable levels of packaging and packing (see 5.1).

d. Federal Stock No. or other identification number.

e. Whether sampling plan B test should be omitted (see 4.3.2.2).

* 6.3 Design approval. Newly designed or design changes to clamps not previously furnished under this specification are subject to approval of the responsible engineering activity.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AV
Navy - AS
Air Force - 82

Review Activities:
Army - AV
Navy - AS
Air Force - 82
DSA - IS

Preparing Activity:
Air Force - 82

Project No. 5340-0921

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:30Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:30Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

DEPARTMENT OF THE AIR FORCE

OFFICIAL BUSINESS

SAN ANTONIO AIR MATERIEL AREA
SERVICE ENGINEERING DIVISION (MMEOP)
KELLY AFB, TEXAS 78241

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:30Z
Check the source to verify that this is the current version before use.